SIDDESH PATIL

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SUMMARY

Computer Science & Engineering (Data Science) graduate with Honours in Artificial Intelligence and Machine Learning (AI/ML). Published IEEE researcher in AIoT-based precision agriculture. Built end-to-end machine learning projects, including 92.86%-accurate rainfall prediction, 97.2%-accurate crop classification, and a hybrid Retrieval-Augmented Generation (RAG) pipeline. Seeking Machine Learning Engineer, AI Engineer, or Data Scientist roles.

EDUCATION

University of Mumbai 2022 – 2026
Bachelor of Engineering (BE) in Computer Science and Engineering (Data Science), Honours in AI/ML

PROJECTS

SoilStream | *Python, scikit-learn, TensorFlow, ResNet50, ESP32, Flask, Firebase* github.com/triumph10/SoilStream

- Predicted regional rainfall with **92.86% accuracy** using Support Vector Regression (SVR), outperforming LightGBM and GRU baselines, to support irrigation timing during monsoon season.
- Developed a ResNet50 crop-classification model (**97.2% accuracy**) with TensorFlow and integrated it into an automated irrigation controller, reducing waterlogging risk and manual monitoring effort.
- Designed an IoT-to-cloud pipeline (ESP32 + BME280 sensors → Firebase → mobile app) for real-time field monitoring, remote pump control, and crop disease diagnosis via the Kindwise API.

VeriDoc | *Python, FAISS, BM25, Sentence-Transformers, Mistral, MLflow, RAGAS, Gradio* github.com/triumph10/Veridoc

- Engineered a **hybrid retrieval RAG pipeline** over local PDFs, combining dense FAISS vector embeddings with sparse BM25 search, and generated cited answers with the Mistral large language model (LLM).
- Evaluated retrieval and answer quality with the **RAGAS** framework, tracking faithfulness and relevancy metrics across experiment runs in MLflow.

Sidekik (Freelancing Platform) | *Flask, BERT, FAISS, Elasticsearch, Supabase, PostgreSQL*
github.com/triumph10/Sidekik

- Developed a **semantic matching engine** using BERT embeddings and FAISS to rank freelancer-project fit, improving match relevance over keyword-only search across freelancer profiles and project records.
- Implemented semantic search with Elasticsearch and integrated the Adzuna API for real-time skill-demand analytics.
- Delivered a full-stack platform end to end, from data pipeline to deployed product, using Flask APIs, a Supabase (PostgreSQL) backend with vector search, and a marketplace UI.

PUBLICATIONS

Research Paper - IEEE CONIT 2026 (Published) Paper ID: 960
A Solar-Powered AIoT-Based System for Intelligent Soil Monitoring and Automated Irrigation Control

SKILLS

Languages: Python, SQL, Java, C
AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Computer Vision, Convolutional Neural Networks (CNNs), LSTM, GRU, Support Vector Regression (SVR), Recommendation Systems, Semantic Search, Vector Databases, Model Evaluation, Exploratory Data Analysis (EDA), Time Series Forecasting
Libraries & Frameworks: TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Sentence-Transformers, FAISS, LightGBM, LangGraph, Flask, FastAPI, Gradio
MLOps & Tools: MLflow, RAGAS, Docker, Git, GitHub, Jupyter Notebook, Linux
Cloud & Databases: AWS (EC2, S3, IAM), Firebase, Supabase, PostgreSQL, Elasticsearch, REST APIs

CERTIFICATIONS

Generative AI | *Google Cloud Virtual Internship (AICTE)*

Google AI/ML | *Google Cloud Virtual Internship (AICTE)*

Data Science Methodology | *IBM Developer Skills Network*